

A Pre-Analysis Plan for Pediatric Emergency
Care Utilization Patterns by Maternal Mental
Health Status:
*The Role of Behavioral and Family Factors
(BFF)*

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Motivation & Background



1) Pediatric ED Use is a Policy Concern

- ~13% of pediatric ED visits are **preventable**
- Signals **gaps in primary care** access and **social inequities**
- Contributes to **costs** and **ED overcrowding**



2) Poor Maternal MH May Play a Role

- **Linked to higher ED use** among children
- Mechanisms include:
 - **At-home management** capacity
 - Perceived **illness severity**
 - **Access/navigation** barriers



3) Gaps in the Literature

- **Small or high-risk** samples
- Primary focus on **infants (0-1 year)**
- **Limited generalizability** to broader pediatric populations



4) Theoretical Background

- **Family Systems Theory:** parent health affects child outcomes
- **Levesque's Conceptual Framework of Access to Healthcare:** caregiver capacity shapes healthcare use



5) Our Contribution

- Nationally representative **NSCH (2022-2023)**
- Broader **pediatric age range (0-17 years)**
- Maternal MH --> **ED visit frequency**



Research Question



***Among children aged 0-17 years in the US,
what is the estimated causal effect of
maternal MH status on pediatric
ED utilization in the past 12 months?***

Overall Objective: To estimate the **causal effect of maternal mental health on pediatric emergency department (ED) utilization** using observational data from the NSCH.

Aim 1: Estimate the ATE

- To estimate the average causal effect of poorer maternal MH (Fair/Poor vs. Excellent/Very Good/Good) on pediatric ED utilization.
- *Are children whose mothers report worse MH more likely to use the ED, on average?*

Aim 2: Examine HTE via CATEs

- To estimate CATEs to assess whether the effect of maternal MH on pediatric ED utilization varies across key subgroups.
- *Does the effect of maternal MH differ depending on which children or families we're talking about?*

Aim 3: Assess Robustness and Validity of Causal Estimates

- To evaluate the robustness of the estimated treatment effects, conduct a series of diagnostic and sensitivity analyses.
- *Are the results consistent across reasonable modeling choices, or do they depend heavily on specific assumptions?*

Approach/Methods

Data Source

- **NSCH** 2022-2023
- Nationally representative, cross-sectional, **caregiver-reported** data on children ages 0-17
- Captures key indicators related to **healthcare access**, **family characteristics**, and other **SDOH**

Study Design

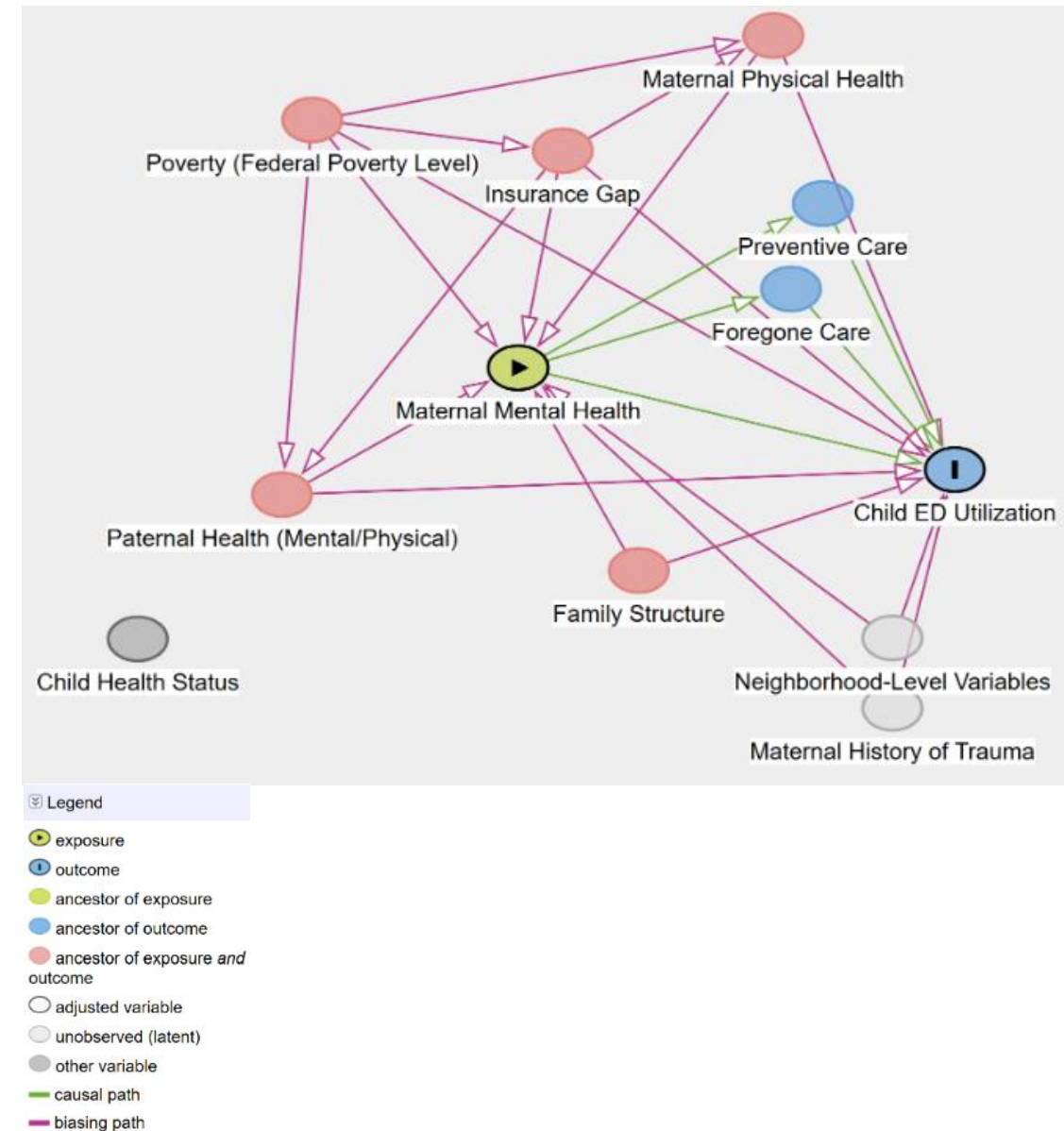
- **Observational** study estimating **causal effects**

Variables of Interest

- **Primary exposure:** Maternal MH (Fair/Poor vs Excellent/Very Good/Good)
- **Primary outcome:** Pediatric ED utilization (0, 1, 2-3, 4+ visits)
- **Core covariates:**
 - Household socioeconomic status (FPL categories)
 - Family structure (e.g., two-parent vs single-parent)
 - Insurance instability (e.g., gaps in coverage)
 - Maternal PH
 - Paternal MH
 - Paternal PH

NSCH=National Survey of Children's Health;
SDOH=social determinants of health; MH=mental
health; ED=emergency department; FPL=federal
poverty level; PH=physical health

DAG



Estimation Strategy



Why DML?

- Observational data --> **risk of confounding**
- Use DML to **flexibly adjust** for covariates and **reduce bias**



What DML is Doing

- Estimate outcome model (**predicts ED visits**) and treatment model (**propensity score**)
- Use **cross-fitting** to avoid overfitting
- Combine both to **estimate ATE**



HTE/CATE

- Estimate CATEs to **assess heterogeneity**
- **Subgroups** include child health status, SES (FPL), and family structure



Sensitivity/Robustness

- **Assess overlap** using propensity scores
- **Compare:** binary vs ordinal outcomes and alternative covariate sets
- Restrict to **common support**

Expected Results

PRIMARY: Children of mothers with poorer MH will have higher ED utilization

Consistent with the primary hypothesis, we expect a **positive relationship** in which **poorer MH** is associated with **increased** reliance on **emergency care** among children.

SECONDARY

Stronger effects among children in poorer health

Greater vulnerability may **amplify** the impact of **maternal MH**, leading to **higher ED utilization** among children.

Larger effects in lower-income households

Reflecting structural **disparities in access** to preventive care, which may **amplify** the impact of **maternal MH** on healthcare-seeking behaviors.

Differences by family structure (e.g., single-parent households)

Variation in **caregiving capacity and support** may modify the relationship b/wn **maternal MH** and **ED utilization**.

Limitations



Unmeasured factors (e.g., parenting style, parent risk perception)



Temporal relationship (e.g., what comes first, ED visit or MH problems? Are they cyclical?)

Questions

- **Thoughts or suggestions for us?**
- **Questions about our approach?**
 - Should we include child health status in secondary or sensitivity analyses, given that it may act as both a mediator and a modifier of the relationship between maternal mental health and ED use?
 - To what extent should we continue investing effort in simulated components of the project, such as the power analysis and resolving `renv::restore()` issues?
 - Are there any concerns about sample selection or missing data in NSCH?

